

MTE 203 – Advanced Calculus

Week 2 Homework Solutions

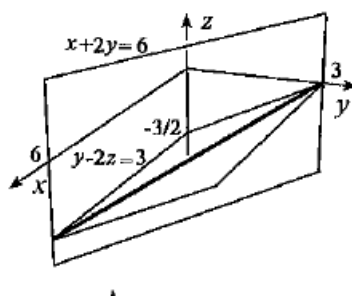
Curves and Surfaces (S.11.2)

Problem 1 [S. 11.2, 37]:

Draw the curve defined by the equations $x + 2y = 6$, $y - 2z = 3$

Solution:

These are the equations of 2 planes and their intersection is a line.



To help us draw the line, we find points of intersection in the $x - y$ plane and the $x - z$ plane using the equations of the 2 planes as,

$$x + 2y = 6 \quad (1)$$

$$y - 2z = 3 \rightarrow y = 2z + 3 \quad (2)$$

Replacing (2) in (1),

$$x + 2(2z + 3) = 6$$

$$x = -4z \quad (3)$$

From (2) and (3), we find the point of intersection with the $x - y$ plane ($z = 0$),

$$(0, 3, 0)$$

and the point of intersection with the $x - z$ plane ($y = 0$),

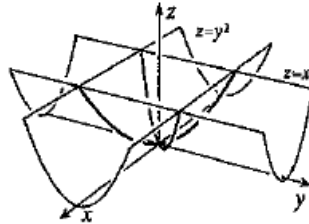
$$\left(6, 0, -\frac{3}{2}\right)$$

Problem 2 [S11.2, 45]:

Draw the curve defined by the equations $z = x^2$, $z = y^2$

Solution:

The given equations represent two parabolic cylinders. One opening along the y -axis and one opening along the x axis.



To aid visualization, it is often helpful to first draw the projections of the cylinders onto the coordinate planes. The projection of the cylinder $z = x^2$ can be easily drawn in the $x - z$ plane, as the unconstrained variable for this surface is " y ". Similarly, the projection of the cylinder $z = y^2$ can be easily drawn in the $y - z$ plane, as the unconstrained variable for this surface is " x ".

Problem 3 [S.11.2, 49] :

Find the equation for the projection in the $x - y$, $y - z$, and $x - z$ planes and draw the curve given by the following equations.

$$x^2 + y^2 = 4, \quad y = x$$

Solution:

The projections in the $x - y$ plane is the line $y = x$. The intersection points can be found by equating the two surfaces,

$$x^2 + x^2 = 4$$

$$x = \pm\sqrt{2}$$

$$y^2 + y^2 = 4$$

$$y = \pm\sqrt{2}$$

These points are therefore,

$$(+\sqrt{2}, +\sqrt{2}) \text{ and } (-\sqrt{2}, -\sqrt{2})$$

The projections in the $y - z$ plane can be found when $x = 0$, which gives us the trivial solution.

Since $y = x$, the projections in the $y - z$ plane are the lines defined by the equation,

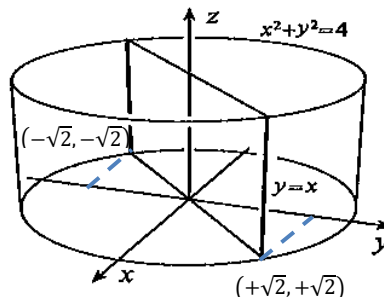
$$2y^2 = 4 \rightarrow y = \pm\sqrt{2}$$

Similarly, the projections in the $x - z$ plane can

be found when $y = 0$. Since $y = x$, the projections in the $x - z$ plane are the two lines defined by the equation,

$$2x^2 = 4$$

$$x = \pm\sqrt{2}$$



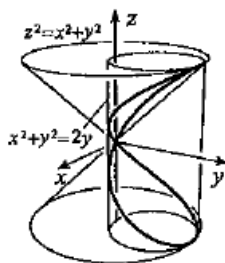
Problem 4 [S.11.2, 61] :

Find the equation for the projection in the x - z plane and draw the curve given by the following equations.

$$x^2 + y^2 - 2y = 0, \quad z^2 = x^2 + y^2$$

Solution:

Drawing of the curve of intersection between the two surfaces,



To find the projection in the x - z plane, we can re-arrange the cylinder equation (equation on the left) as follows,

$$x^2 + y^2 = 2y$$

From the cone equation (equation on the right), we know that $x^2 + y^2 = z^2$, hence replacing this value in the equation above,

$$z^2 = 2y$$

$$y = \frac{z^2}{2}$$

Substituting back in the first equation,

$$x^2 + \frac{z^4}{4} - z^2 = 0$$

$$4x^2 + z^4 - 4z^2 = 0$$

$$(z^2 - 2)^2 = 4 - 4x^2$$

Hence the projection of the curve in the x-z plane is given by,

$$z^2_{1,2} = 2 \pm 2\sqrt{1 - x^2}, \quad y = 0$$

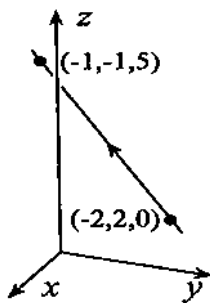
Parametric and Vector Representations of Curves (S.11.10)

Problem 1 [11.10; Prob. 13]

Draw and plot the curve with the given position vector,

$$\vec{r}(t) = (t - 2)\hat{i} + (2 - 3t)\hat{j} + 5t\hat{k}$$

Solution:



Re-writing the given equation as,

$$\vec{r}(t) = (-2, 2, 0) + t(1, -3, 5)$$

This is a straight line through the points $(-2, 2, 0)$ and $(-1, -1, 5)$, where $t \in \mathbb{R}$ (or $-\infty < t < \infty$).

Problem 2 [11.10; Prob. 15]

Draw and plot the curve with the given position vector,

$$\vec{r}(t) = (\cos t)\hat{i} + (\sin t)\hat{j} + (\cos t)\hat{k}, \quad 0 \leq t \leq \pi$$

Solution:

If we set,

$$x = \cos t$$

$$y = \sin t$$

Then,

$$z = x$$

Forgetting about z for a moment, we can see that

$$x^2 + y^2 = 1$$

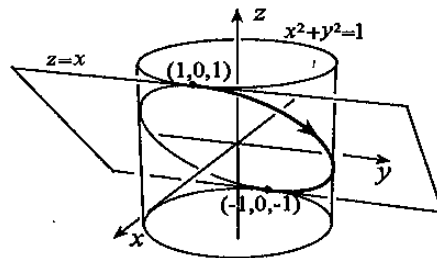
Which is a half cylinder (due to the range of t), running along the whole z axis.

The equation $z = x$, is a 45-degree slanted plane. To draw the intersection of the plane with the half cylinder, let us determine some points of intersection between them:

If $y = 0$: $(1,0,1)$ and $(-1,0,-1)$

If $x = 0$: $(0,1,0)$ and $(0,-1,0)$

Therefore, the intersection of these two surfaces can be drawn as,

**Extra Practice Problems¹**

The answers to these problems, along with all even-numbered exercises, are provided at the back of the textbook. Detailed solutions can be found in Trim's Student Solution Manual. If you don't own a copy, a reserved edition is available at the Davis Centre.

1. S. 11.2, Probs. 14, 44, 54, 68
2. S. 11.10, Probs. 14

¹ You are strongly encouraged to try and solve the problems assigned in this homework on your own, before looking at the solutions. These problems are only the minimum necessary to master the material. You are also encouraged to do additional problems from the text for practice (some suggested **extra practice problems** have been listed here).